

Halogen bonds as orthogonal molecular interactions to hydrogen bonds

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Halogen bonds (X-bonds) are shown to be geometrically perpendicular to and energetically independent of hydrogen bonds (H-bonds) that share a common carbonyl oxygen acceptor. This orthogonal relationship is accommodated by the in-plane and out-of-plane electronegative potentials of the oxygen, which are differentially populated by H- and X-bonds. Furthermore, the local conformation of a peptide helps to define the geometry of the H-bond and thus the oxygen surface that is accessible for X-bonding. These electrostatic and steric forces conspire to impose a strong preference for the orthogonal geometry of X- and H-bonds. Thus, the optimum geometry of an X-bond can be predicted from the pattern of H-bonds in a folded protein, enabling X-bonds to be introduced to improve ligand affinities without disrupting these structurally important interactions. This concept of orthogonal molecular interactions can be exploited for the rational design of halogenated ligands as inhibitors and drugs, and in biomolecular engineering.

Orthogonality is a well-recognized concept in mathematics and other sciences, relating things that are physically perpendicular or functionally independent of each other. In chemistry, for example, orthogonal protecting groups can be added and removed independently of each other, and therefore are commonly used to control the specificity of reactions in the synthesis of complex macromolecules, including polypeptides and polynucleotides¹. There are many examples of individual molecular interactions that can be considered to be geometrically orthogonal to chemical functional groups: for instance, the aromatic side-chains in proteins prefer perpendicular edge-to-face geometries^{2,3}, whereas electronegative groups tend to approach an electrophilic amide carbon perpendicularly to the molecular plane⁴. Pairs of molecular interactions, however, are not usually considered to be orthogonal: hydrogen bonds, dipole–dipole, steric and solvent interactions will typically either complement or compete with each other to define the folding of complex molecules, including proteins, DNA and RNA, as well as ligand binding affinities and specificities to proteins. We propose here that halogen bonds (X-bonds) in protein–ligand complexes are both geometrically perpendicular to and energetically independent of classic hydrogen bonds (H-bonds), suggesting that they can be considered to be orthogonal molecular interactions. This concept is distinct from other ‘orthogonal interactions’ in that it provides a strategy for predicting the optimum geometry for engineering X-bonds to control macromolecular structure and ligand binding in proteins through rational approaches⁵ without disrupting pre-existing H-bonds.

H-bonds are well recognized as molecular interactions that stabilize the structures of α -helices and β -sheets in proteins, and the helices of DNA and RNA. X-bonds, once known as ‘charge-transfer bonds’⁶, are less widely known, but share many common properties with H-bonds: both can be treated as directional electrostatic interactions between Lewis acids and bases, although H-bonds are less directional than X-bonds⁷. The less electronegative hydrogen, relative to its heavy donor atom (denoted by ‘D’ in Fig. 1a), allows it to serve as the Lewis acid in an H-bond. In an X-bond, the electropositive crown¹⁰ that results from polarization of the halogen (Cl, Br or I) along its covalent bond^{11–13}, referred to as a σ -hole¹⁴, is the Lewis acid donor. Oxygens, nitrogens and sulfurs

can serve as the Lewis base acceptors in both interactions (Fig. 1a). The similarities in the chemical and physical properties of H- and X-bonds suggest that they would compete for the same interaction sites of an acceptor, which they do in many organic complexes⁵. Indeed, such competition can be used to control crystallization of simple organic compounds for the synthesis of new materials¹⁵.

In biological molecules, the situation is more complex, because of the multitude of interactions that contribute to their three-dimensional structures. A survey of protein and nucleic acid crystal structures showed one group of X-bonds that are similar to those in organic complexes in their interactions with the non-bonding electron pairs of oxygens in hydroxyl groups, peptide bonds and phosphodiester bonds¹³. These X-bonds, as with their small-molecule counterparts, can be placed in competition with a hydrogen bond to drive biomolecular conformation, for example in directing the isomerization of a four-stranded DNA Holliday junction¹⁶. A second group of X-bonds, representing the majority of interactions of ligands with proteins, differ significantly in that the halogen shares its acceptor oxygen with, rather than competing against, an H-bond. It is this large and general class of X-bonds (designated as *hX*-bonds) that we demonstrate here to be orthogonal to the classic H-bond in proteins. To support this concept, we first demonstrate that *hX*-bonds are geometrically perpendicular to the shared H-bonds in the structures of protein–ligand complexes. This spatial relationship is shown, through *ab initio* calculations, to arise from the distribution of electrostatic potentials at the oxygen surface. Finally, *ab initio* calculations on simple model systems demonstrate that the presence of an *hX*-bond does not affect the energy of an established H-bond, and vice versa. This concept of orthogonal molecular interactions will allow us to predict the optimum geometry of X-bonds that would not perturb the H-bonds that establish the structure and stability of a natively folded protein.

Results

For the current study, we started by compiling a dataset of crystal structures of protein–ligand complexes that show X-bond interactions of any type. Nearly all of the X-bonds within this dataset

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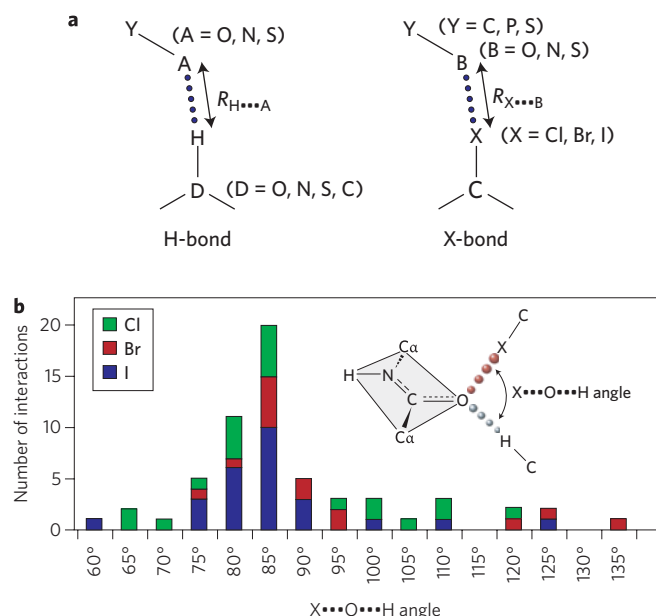


Figure 1 | H- and X-bond geometries. **a**, Comparison of an H-bond with an X-bond interaction with a Lewis base acceptor (labelled A for the H-bond and B for the X-bond). The interactions are characterized by a hydrogen-to-acceptor distance $R_{H...A}$ or halogen-to-base distance $R_{X...B}$ that is shorter than the sum of their respective van der Waals radii. **b**, Histogram of the relative angle of approach of H- and X-bonds to a shared Lewis base (oxygen) for each type of X-bond (Cl, Br or I) observed in structures in the Protein Databank²⁵.

involve a carbonyl oxygen that is also H-bonded (including H-bonds from other peptide bonds, from amino acid side chains or from water; Table 1). We designate the set of ligand–protein structures in which H- and X-bonds share a common peptide oxygen the *hX*-dataset. The *hX*-bonds in the *hX*-dataset are seen to be narrowly distributed around an average angle of 88.2° (s.d. = 14.1°) relative to the shared H-bond (Fig. 1b).

To explore the geometric relationship between X- and H-bonds in greater detail, we treated the carbonyl oxygen of the peptide bond in the *hX*-dataset structures as a reference point, and defined for its surfaces a set of azimuthal angles (a and b) to relate the *hX*- and H-bond interactions on a latitudinal/longitudinal grid (Fig. 2a). H-bonds are seen to populate primarily the equatorial regions (Fig. 2b), whereas *hX*-bonds tend to populate the polar regions (Fig. 2c) of the oxygen surface, with very little overlap between the two regions. Not all *hX*-bonds, however, are at the polar regions; nearly 15% lie near the equator, but remain perpendicular to their shared H-bonds. In these cases, the H-bonds either sit at a polar region, or near the equator, but with b -angles that maintain the perpendicular geometry. It is clear, therefore, that *hX*-bonds are geometrically orthogonal to H-bonds when they share a common Lewis base acceptor.

As both H- and *hX*-bonds are primarily electrostatic interactions, we would expect that their geometries would be defined by the electrostatic properties of the oxygen acceptor. To test this assumption, we calculated the electrostatic potential of the oxygen in the amide bond of *N*-ethylpropanamide (NEPA, 1) (Fig. 3a), a simple molecular model for a peptide bond within a polypeptide chain. The carbonyl oxygen of NEPA shows electronegative potentials both in-plane and out-of-plane of the amide bond (Fig. 3b), with the orientation of the out-of-plane potential dependent on the rotation of the neighbouring C_α and C_β carbons. These are equivalent to the ϕ -rotation (about the amide nitrogen and C_α carbon) and ψ -rotation (about the C_α and carbonyl carbons) of the peptide bonds along a protein chain. The X-bonding energy calculated for a bromobenzene (2) molecule to the NEPA oxygen centre mirrors the distribution of electronegative potentials across the oxygen surface (Fig. 3c), with highly negative interaction energies seen both in and out of the peptide plane. The region at b -angle around $+60^\circ$, although highly electronegative, has very large positive X-bonding energies because of the steric clash between the brominated ligand and the adjoining C_α and C_β carbons. This map of X-bonding interaction energies also corresponds well to the distribution of *hX*-bonds across the oxygen surface (Fig. 2c) as seen in the *hX*-dataset, confirming that NEPA is an appropriate model for studying X-bonds to internal peptide bonds in proteins. These results indicate that amide oxygens, in the context of a polypeptide chain, would present two perpendicular electronegative interaction surfaces to potential H- or X-bond donors, providing the necessary electrostatic foundation for their perpendicular geometries.

The effect that protein structure has on the relationship between *hX*- and H-bonds can be seen by comparing their geometries in α -helices and β -sheets. These two classes of protein secondary structures are characterized by well-defined sets of ϕ - ψ rotations and are therefore expected to show different distributions of *hX*-bonds, particularly across the a -angles that distinguish between equatorial and polar interactions to the oxygen surface. The distribution of a -angles of *hX*-bonds for β -sheets is bimodal, with peaks at or near $\pm 90^\circ$ ($\langle |a| \rangle = 82.5^\circ \pm 6^\circ$) (Fig. 4a). The reason for this is fairly simple: the H-bonds holding the strands of a β -sheet together are essentially in the plane of the peptide bonds and the resulting sheet. Thus, *hX*-bonds to these shared oxygens are constrained to approach from above or below the sheet (Fig. 4b).

In α -helices, however, the a -angle distribution shows a single strong peak centred at -50° ($\langle a \rangle = -49.4^\circ \pm 20^\circ$), but spanning the full range from -90° to $+90^\circ$ (Fig. 4c). The rationale for this distribution is not as obvious, and may suggest that the concept of orthogonality breaks down for helices. A more detailed analysis, however, shows that the amino hydrogen in an ideal helix is $+38^\circ$ out of plane relative to the $O=C$ bond of its H-bonded partner (Fig. 4d). Thus, although the mean a -angle for the X-bond is -50° in terms of the peptide plane, it remains geometrically perpendicular to the associated H-bonded amino hydrogen. The few *hX*-bonds with positive a -angles are attributed to interactions

Table 1 | Number of X-bonds that share a common oxygen acceptor with an H-bond.

X-bond halogen	Number of X-bonds	Number of associated H-bond interactions				Average $\angle(X...O...H)$ in <i>hX</i> -dataset
		Peptide N–H	Protein side chain	Water	None	
Cl	24	10	3	7	4	88.3° (15.5°)
Br	21	16	1	0	4	95.0° (18.2°)
I	31	27	0	2	2	83.9° (9.7°)
Total	76	53	4	9	10	88.2° (14.1°)

The numbers of X-bonding interactions involving chlorine-, bromine- or iodine-type halogens with a peptide oxygen in proteins are grouped into those that share an oxygen with an H-bonding interaction—from the amino group of another peptide bond, from a protein side chain or from a solvent molecule (the *hX*-dataset)—and those that show no associated H-bond. The average angles and the standard deviations (in parentheses) relating the *hX*- and H-bonds in the *hX*-dataset are listed for each type of halogen.

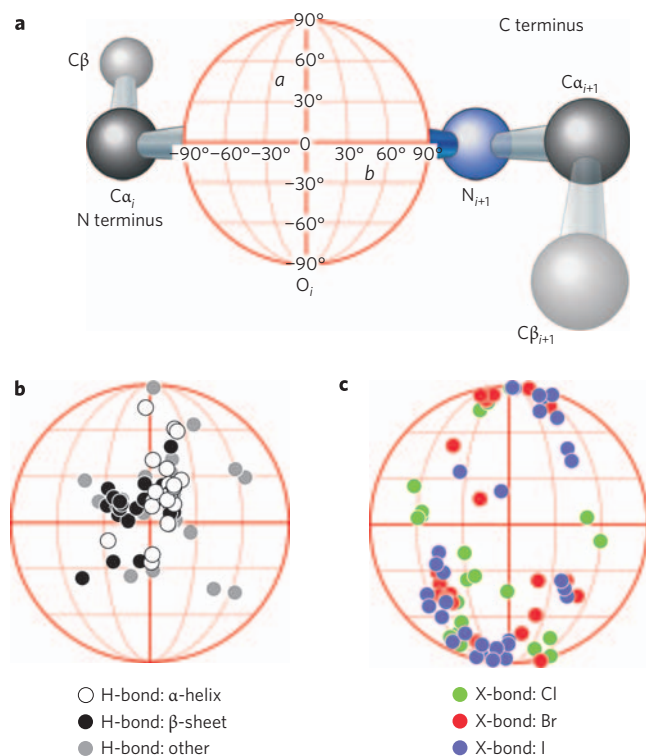


Figure 2 | Distribution of H- and hX-bonds at the carbonyl oxygen surface in proteins. **a**, Definition of latitudinal (a) and longitudinal (b) azimuthal angles of the oxygen surface. The oxygen is viewed down the O=C bond, with the N terminus of the polypeptide chain to the left and the C terminus to the right. In this reference frame, positive a -angles are above the peptide plane and positive b -angles to the right (towards the C terminus). **b**, Distribution of H-bonds across the oxygen surface. White filled circles represent H-bonds in α -helices, black circles those in β -sheets, and grey circles those for other structures. **c**, Distribution of X-bonds across the oxygen surface. Points of contact for chlorine X-bonds are shown in green, those for bromine in red and those for iodine in blue.

at the C termini of helical segments, where H-bonds are not geometrically constrained by the helix.

One of the complexities in protein structures is the strong effect of steric interactions in defining the native conformation. To study these effects on the orthogonal relationship between hX- and H-bonds, we compared the accessible surface of the carbonyl oxygen atoms in an ideal β -sheet and an ideal α -helix. We see that the oxygen surfaces on either side of the β -sheet are accessible to probes with radii up to 2.2 Å, and that these exposed surfaces lie approximately parallel to the plane of the H-bonded peptide bonds across strands (Fig. 4b). Bromine atoms positioned in the centres of these accessible surfaces sit approximately perpendicular to the peptide plane, consistent with the distributions of hX-bond a -angles observed to β -sheets.

In the α -helix, the oxygen surface is accessible to a larger range of probe radii (up to 3.4 Å); however, only the face of the oxygen associated with negative a -angles is accessible. In this case, the centre of the accessible surface is not parallel to either of the two H-bonded peptides, but sits at an angle of about 50° relative to the carbonyl group of the H-bonded peptides (Fig. 4d). The conformation of the helix places a C β carbon (which represents the amino acid side-chain in a protein) just above the peptide plane, restricting access to the oxygen from the carbon side of the O=C carbonyl bond. Thus, there is a strong steric contribution to the a -angle distributions of both the α -helices and β -sheets.

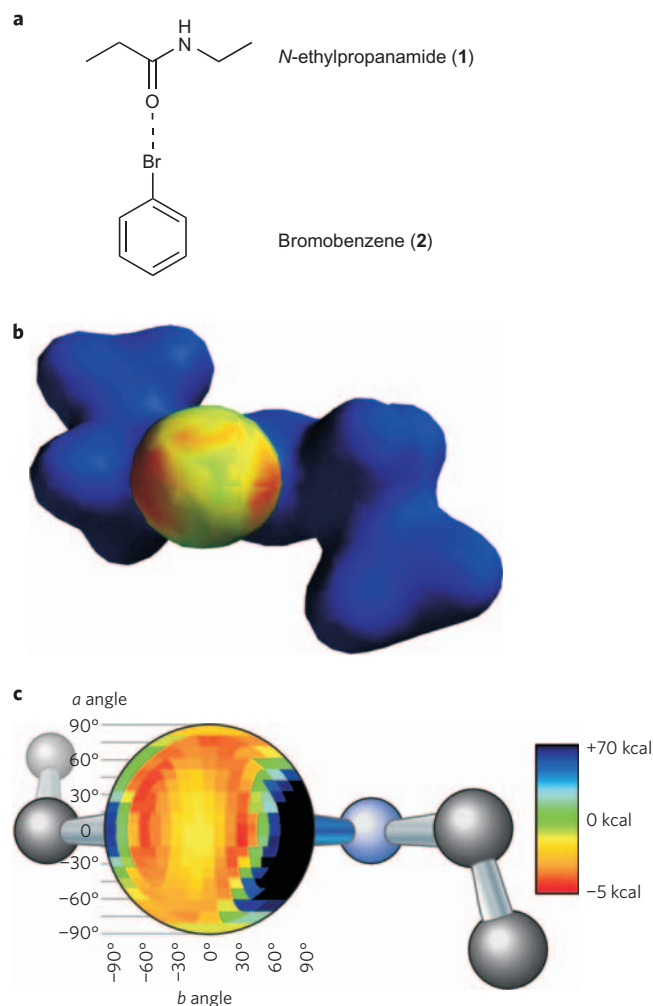


Figure 3 | Electrostatic potentials and X-bonding interaction energies of the carbonyl oxygen in NEPA (1). **a**, Chemical structures of **1** and bromobenzene (**2**) model compounds in an X-bonding geometry. **b**, Electrostatic potentials from *ab initio* calculations mapped onto the molecular surface of NEPA and visualized with SPARTAN. The carbonyl oxygen is oriented towards the viewer. The most electronegative potentials are shown in red; neutral and electropositive potentials are shown in blue. **c**, X-bond interaction energies calculated for bromobenzene to the carbonyl oxygen of the NEPA. A bromobenzene model was placed with the halogen 2.8 Å (see Fig. 5b) from the centre of the carbonyl oxygen, and rotated about the oxygen centre (with the benzene ring oriented away from the oxygen centre) to sample azimuthal angles from $\pm 90^\circ$ (a -angles labelled vertically and b -angles labelled horizontally). The *ab initio* energies calculated for each set of a - and b -angles were mapped onto the oxygen surface, with red representing interaction energies of -5 kcal, green 0 kcal and black +70 kcal (see the key in the side bar).

To determine whether hX-bonds are energetically independent of their shared H-bonds, we calculated the *ab initio* energies of an H-bond between two NEPAs in the presence and absence of an hX-bonded bromobenzene (Fig. 5a), and, similarly, the energy of an X-bonded bromobenzene with a NEPA molecule in the presence and absence of a second H-bonded NEPA (Fig. 5b). The magnitudes of the energies for H- and X-bonding interactions calculated here are similar to those from previous *ab initio* calculations^{17,18}. The current results show that introducing an hX-bond has basically no effect on the interaction energies of an H-bond to the same carbonyl oxygen, and vice versa, indicating that orthogonality extends to the physico-chemical as well as stereochemical relationships of the two interactions.

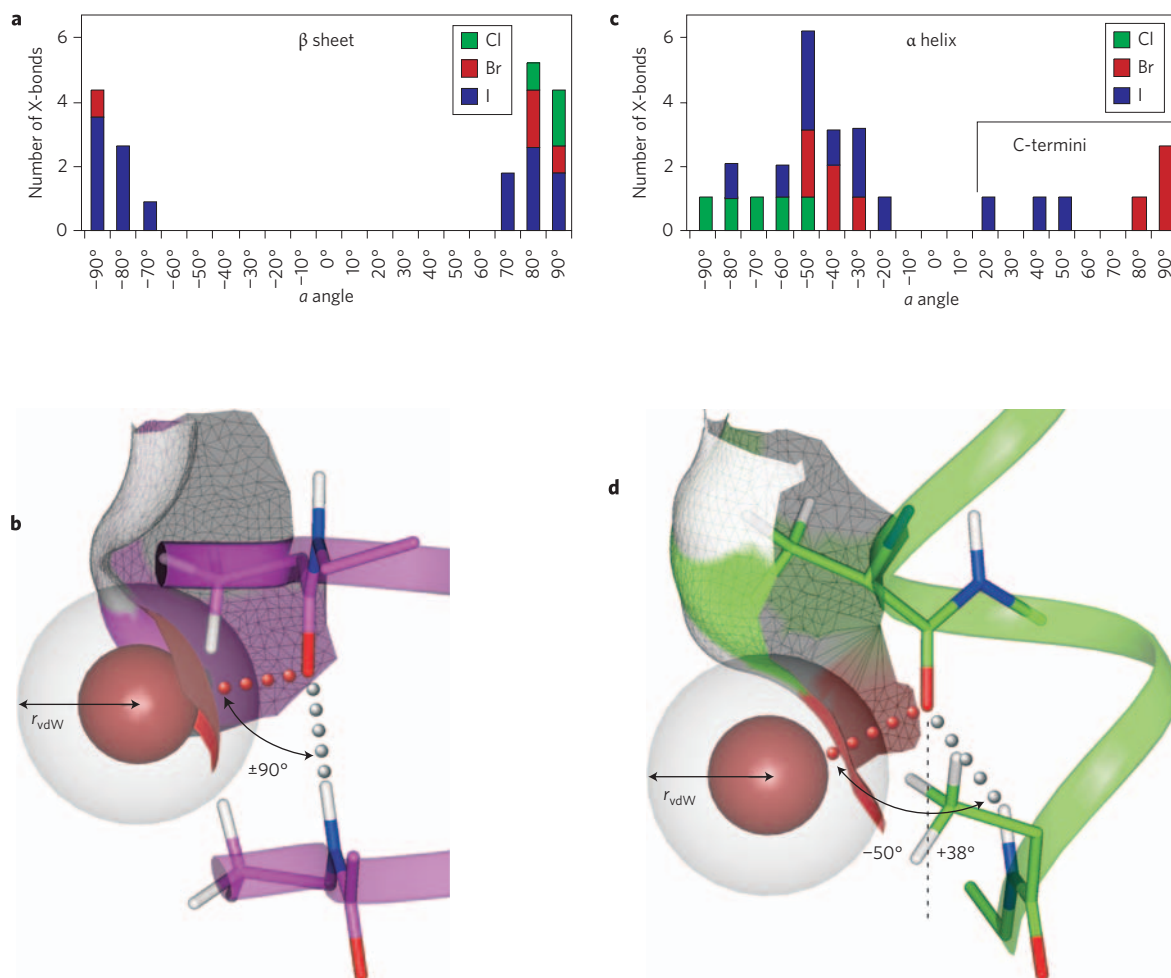


Figure 4 | X-bond geometries in β -sheets and α -helices. **a**, Histogram of the α -angles for X-bonds to peptide oxygens of β -sheets. **b**, The accessible surface (calculated with a 2.2-Å-radius probe) of an ideal β -sheet forms a pocket that accommodates a halogen atom. A Br atom placed 2.8 Å from the carbonyl oxygen is shown with half its van der Waals radius (red sphere) sitting in this pocket, and the full van der Waals radius (white sphere) impinging into the accessible surface of the peptide oxygen in order to form the X-bond. This pocket forms at an α -angle of $\pm 90^\circ$, either above or below the plane of the peptide, and is orthogonal to the H-bond between the two peptides. **c**, Histogram of the α -angles for X-bonds to the peptide oxygens of α -helices. **d**, A pocket is formed in the accessible surface (calculated with a 3.4-Å probe) at the carbonyl oxygen of the peptide bond of an ideal α -helix. A Br atom placed 2.8 Å from the carbonyl oxygen is shown with half its van der Waals radius (red sphere) sitting in this pocket, and the full van der Waals radius (white sphere) impinging into the accessible surface of the peptide oxygen to form the X-bond. The centre of this pocket sits approximately 50° relative to the peptide plane and perpendicular to the hydrogen of the H-bond that is aligned approximately 38° to this oxygen. α angles $> 0^\circ$ are associated with X-bonds at the C termini of helices.

This concept of orthogonality is not, however, general to all molecular interactions with peptide oxygens. A similar analysis (Fig. 5c) shows that an H-bonded water molecule significantly perturbs the energy of interaction between the two H-bonded NEPA peptides ($\Delta E \approx 1.5 \text{ kcal mol}^{-1}$, or nearly 20%), underscoring the exceptional nature of the thermodynamic independence between the hX - and H-bonds. Similarly, the geometries of double H-bonds, where two hydrogen donors share a common acceptor (as opposed to the better-known bifurcated H-bond, where a single donor is shared between two acceptors), are not as obviously perpendicular or as stringently defined¹⁹ as the orthogonal hX - and H-bonds. The *ab initio* calculations suggest that such double H-bonds are less favourable than the sum of two isolated H-bonds, which would be consistent with the observed longer donor-acceptor distances in double versus single H-bonds in proteins¹⁹.

Discussion

In protein-ligand complexes, hX -bonds are seen to be geometrically perpendicular to and relatively energetically independent of

H-bonds that share a common acceptor, suggesting that they are uniquely orthogonal molecular interactions—our analyses provide the rationale for both components of this concept. The geometric relationship is established by the in-plane and out-of-plane electronegative potentials of the shared oxygen acceptor. As electrostatic interactions to the non-bonding orbitals are more energetically favoured, H-bonds that help fold a protein into its native conformation form primarily, although not exclusively, along the equatorial plane of the peptide oxygen. A halogenated compound that binds in the ligand pocket of the folded protein can form an hX -bond to a carbonyl oxygen atom that is already H-bonded. The local conformation in the pocket defines the ϕ - and ψ -angles, which, in turn, establish the orientation of the H-bonded amino nitrogen atom, of the electronegative potentials across the oxygen atom's surface, and of the C β carbons that affect the oxygen atom's accessible surfaces. Thus, hX -bonds are orthogonal to H-bonds because of the availability of two sets of relatively independent electronegative potentials that allow for independent electrostatic interactions, the highly directional nature of X-bonds in general, and steric

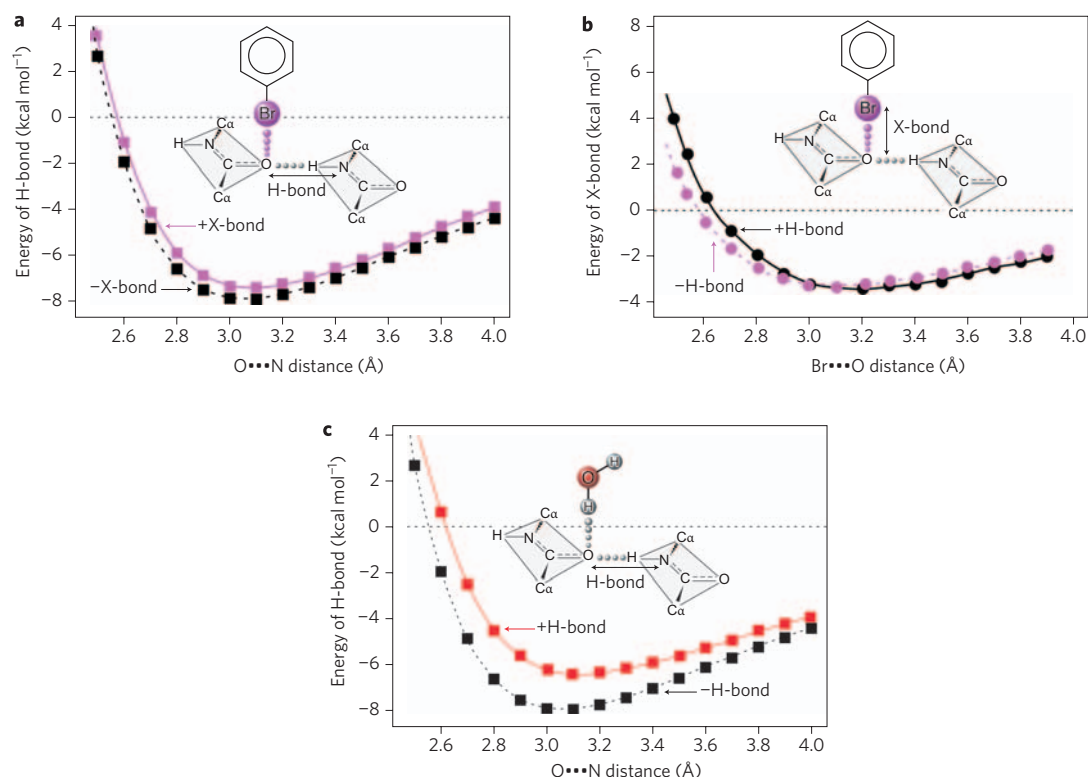


Figure 5 | Effects of X-bonds and H-bonds on interaction energies of NEPA. **a**, Effect of a perpendicular X-bond on in-plane H-bond energies. Two NEPA molecules were placed coplanar to each other, with the C=O bond of one aligned with the H—N bond of the other. *Ab initio* energies for this H-bonding interaction in the absence of an X-bond (black squares and curve) were calculated as that of the H-bonded duplex, minus the energies of the individual NEPA components, at each distance. The H-bond energies in the presence of an X-bond (purple squares and curve) were calculated with this NEPA pair, but with a bromobenzene molecule placed with the bromine 2.8 Å from the H-bonded oxygen and oriented perpendicular to the peptide plane. The H-bond energies were calculated as the energies of the complex minus those of the X-bonded NEPA • bromobenzene complex and the non-X-bonded NEPA molecule, with this latter molecule placed 2.5–4.0 Å (N...O distance) from the X-bonded NEPA. **b**, Effect of an H-bond on the energy of an X-bond. A bromobenzene molecule was aligned perpendicular to the plane, with the bromine towards the carbonyl oxygen of a NEPA molecule. The *ab initio* energies of the X-bond interaction in the absence of an H-bond (purple circles and curve) were calculated as the energy of this complex, minus those of the individual components, as the bromobenzene molecule was moved from 2.5 to 4.0 Å from the NEPA oxygen. The X-bond energies in the presence of an H-bond (black circles and curve) were calculated by placing a second NEPA 2.85 Å (N-to-O distance) from the X-bonded NEPA molecule (see **a**). The X-bonding energies were calculated as that of the trimolecular complex, minus the energies of the H-bonded NEPA pair and the bromobenzene molecule, as the bromobenzene was moved from 2.5 to 4.0 Å from the X-bonded NEPA oxygen. **c**, Effect of a perpendicular H-bond on the interaction energy of an in-plane H-bond. The H-bond interaction energies in the absence (black squares and curve) and presence (red squares and curve) were calculated as described in **a**, except that a water molecule was placed 2.85 Å from the H-bonded NEPA in place of the bromobenzene.

constraints that define the accessible surfaces of the shared oxygen acceptor. Once formed, the *hX*-bond does not perturb the energy of the H-bonds in the folded protein, nor does the H-bond generally affect the energy of the *hX*-bond. This does not mean, however, that X-bonds are entirely independent of H-bonds, as no two molecular interactions can be truly independent of each other. The concept proposed here is that orthogonal X- and H-bonds are analogous to the orthogonal protecting groups in chemical syntheses, which can be added or removed ‘independently’ of each other.

Consequently, this unique orthogonal relationship allows the design of *hX*-bonding interactions that target carbonyl oxygens in a specific orientation relative to and without competing against pre-existing H-bonding interactions, including the structural H-bonds in, for example, α -helices and β -sheets. Thus, we can now start to develop strategies to help design halogenated compounds that will optimize the geometries of *hX*-bonding interactions, particularly in the rational design of ligands and inhibitors for protein drug targets. The results presented here indicate that, starting with the set of H-bonds established in a folded protein structure, an optimum geometry for a potential *hX*-bond can be predicted based on the electronic nature and the accessibility of the H-/X-bond

acceptor, and that such *hX*-bonds can be added or removed without displacing or affecting the interaction energies of these H-bonds. Other potentially ‘orthogonal’ pairs of molecular interactions could include, for example, π – π stacking and H-bonds in aromatic compounds; there is, however, to the best of our knowledge, no literature on the energetic interplay among such pairs of molecular interactions, and it is not clear how such interactions can be applied in an analogous predictive manner. The concept of orthogonal interactions presented here thus serves as the first step towards a rational approach to the design and synthesis of, in this case, halogenated ligands that take advantage of *hX*-bonding as a means to enhance binding and recognition, and that can be exploited to engineer new protein architectures and assemblies⁵ in a manner similar to that in organic supramolecular assemblies^{20,21}.

Methods

The dataset of X-bonds in biological macromolecules was assembled as previously described¹³, and was reduced to represent only structures where X-bonds share carbonyl oxygen atoms with H-bonds (*hX*-dataset). These structures were identified through visual inspection of each X-bond acceptor, and identifying any non-hydrogen H-bond donor that is within 2.6–3.2 Å of this acceptor. For each X- and H-bond interaction, the azimuthal angles (*a* and *b*, in the latitudinal and

longitudinal coordinate system) to the carbonyl oxygen atoms of the peptide bonds were calculated relative to the O=C bond (explicitly defined as $a = b = 0^\circ$) and the peptide plane (defined by the $O_i=C_i-N_{i+1}$ atoms of the peptide bond, for the i th and $(i + 1)$ th residues). Positive a -angles are defined as being above the peptide plane when viewed into the carbonyl oxygen with the N_{i+1} atom to the right, and positive b -angles in the direction of the N_{i+1} atom (Fig. 2a).

Electrostatic potentials were calculated and displayed using the program SPARTAN (Wavefunction, Irvine, California). Interaction energies of *N*-ethylpropanamide (1) with bromobenzene (2) were calculated with the program GAMESS^{22,23} using the WebMO interface (version 6.0), with density functional theory, applying the B3LYP function and the 6-31G(d) basis set. Note, however, that the electrostatic and interaction energy calculations are intended for qualitative comparisons, and should not be considered to be quantitative.

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Author contributions

A.R.V. and P.K. contributed equally to this work. P.S.H., A.R.V. and P.K. analysed and interpreted results, and contributed to writing the manuscript. K.O. performed calculations for this study.

Additional information

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